CI33/CY33

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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – MARCH 2024

Program	:	**B.E. – CSE (Artificial Intelligence and Machine Learning) / CSE (Cyber Security)**	**Semester**	:	**III**
Course Name	:	**Data Structures**	**Max. Marks**	:	**100**
Course Code	:	**CI33 / CY33**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.
- Draw diagram wherever required.

UNIT – I

1.	a)	Define Data Structure. Describe how they are classified. Give example for each.	CO1	(06)
	b)	Develop a C Function REVERSE(P,N) where P is the pointer to an integer array and N is the size of array and reverse the array contents using pointer.	CO1	(07)
	c)	Differentiate between malloc() and calloc() with syntax and example.	CO1	(07)
2.	a)	Develop a 'C' program to implement a structure called Student with members like usn,name,marks1,marks2,marks3 ,total. Write functions to i) Read the details of 'n' students. ii) To find the average of best two marks of each student. iii) Search for the student's details based on usn and display. iv) To display the details of all the students. Use dynamic allocation for structure.	CO1	(10)
	b)	Write a program to create an array of integers by allocating memory dynamically and perform the following operations by writing functions for each operation. i)READ: To read the array elements ii) WRITE: To Display the array elements iii) SEARCHCOUNT which searches for a key value and count its occurrences.	CO1	(10)

UNIT – II

3.	a)	Write a C program to implement circular queue of integers to perform insert, delete and display operations on it.	CO2	(08)
	b)	Illustrate the process of converting the following infix expressions into postfix expressions using stacks: (A+(B*C)/(D-E)+F-G/H).	CO2	(08)
	c)	Differentiate between iteration and recursion. Write a recursive C program to solve "Tower of Hanoi problem".	CO2	(04)
4.	a)	Describe priority queues and list the applications of priority queue.	CO2	(06)
	b)	Write a C program to implement parenthesis checker using stacks.	CO2	(08)
	c)	Write algorithm for evaluation of a postfix expression. Demonstrate the use of this algorithm for evaluating postfix representation of 5+6*7-8.	CO2	(06)

UNIT – III

5.	a)	Implement the following tasks as functions in C for a singly linked list (SLL) with a pointer START pointing to the first node, and an integer is stored in each node as data. Write a menu driven program to demonstrate the use of these functions. i) Traversal of SLL ii) Insert a node at the end of SLL iii) Deletion of a node with given key value.	CO3	(10)
	b)	Write 'C' functions for the following operations on a doubly linked list with a pointer START pointing to the first node. i) Insert a node at start ii) Insert a node after a given key value iii) Insert a node at the end	CO3	(10)
6.	a)	Write the following pseudocode for circular linked list (CLL) with a pointer START pointing to the first node. i) Insert a node start of the CLL ii) Insert a node at the end of CLL iii) Delete a node at the end of CLL.	CO3	(10)
	b)	Write a program to create a singly linked list that stores details of students (Name, USN, CGPA)in the nodes. Write a function that takes the start pointer of the linked list, search and display the details of the student with highest CGPA.	CO3	(10)

UNIT – IV

7.	a)	Explain different traversals in a binary tree with suitable examples.	CO4	(08)
	b)	Construct the binary tree stepwise from its preorder (ABCDFEG) and inorder (BAFDCEG) traversals.	CO4	(06)
	c)	Write a C function swap tree that takes a binary tree and swaps the left and right children of every node.	CO4	(06)
8.	a)	Explore real-world applications of trees in computer science. How are trees used to solve specific problems or represent hierarchical relationships?	CO4	(08)
	b)	Explain each of the following with an example. i) Tree ii) Degree of a tree iii) Height of a tree.	CO4	(06)
	c)	Construct a binary tree for the given inorder and preorder traversals. Inorder traversal : b d h f a e g c Preorder traversal: a b d f h c e g.	CO4	(06)

UNIT- V

9.	a)	For the inputs 50,10,59,75,21,18 and the hash function h(k)=k mod 11, construct the hash tables using open addressing and chaining. Use linear probing for collision resolution.	CO5	(10)
	b)	Write algorithms for insertion and deletion in a B-tree. Demonstrate the algorithms with suitable examples.	CO5	(10)
10.	a)	What is Collision? Discuss various methods to resolve collision? Explain linear probing with example.	CO5	(10)
	b)	Differentiate between B Tree and B+ Tree. Discuss insertion and deletion operation in B+ tree with suitable example.	CO5	(10)

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